Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient's X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient's condition?

Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<thought>

To solve this problem, we need to understand the symptoms and the diagnostic findings. The patient has a non-productive cough, fever, and patchy reticular opacities in the lungs, which are consistent with a community-acquired pneumonia. The presence of these symptoms and the radiological findings point towards a bacterial infection.

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<answer>

C. Peptidoglycan is its major antigen that induces an immune response.</answer>

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The pathogen causing this patient's condition is likely a Gram-negative bacterium. Gram-negative bacteria, such as *Legionella pneumophila*, produce peptidoglycan, which is a major component of the bacterial cell wall. This peptidoglycan is recognized by the immune system and triggers an immune response.

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